HW2

October 21, 2021

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# PHYS472J: Cosmology HW2
# Authored by Oscar Hernandez for Dr. Martin Truffer's Advanced Topics in␣
 ↪Physics II module on Cosmology
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[2]:
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import matplotlib.pyplot as plt
import numpy as np

from math import sqrt, sin
from scipy.constants import pi, c, G
from scipy.integrate import odeint

p = 9*10**(-27) # density of universe in kg/m^3. used to compute␣
 ↪non-dimensionalize R(t) via division by R^*
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## 1--2) Solving the Friedmann Equation of universal expansion

a = 0 # start of non-dimensional time
b = 2*pi # end of non-dimensional time
eta = np.linspace(a, b, 201) # non-dimensional time discretization
n = len(eta) # number of non-dimensional time increments
deta = (b-a)/n # size of non-dimensional time increments

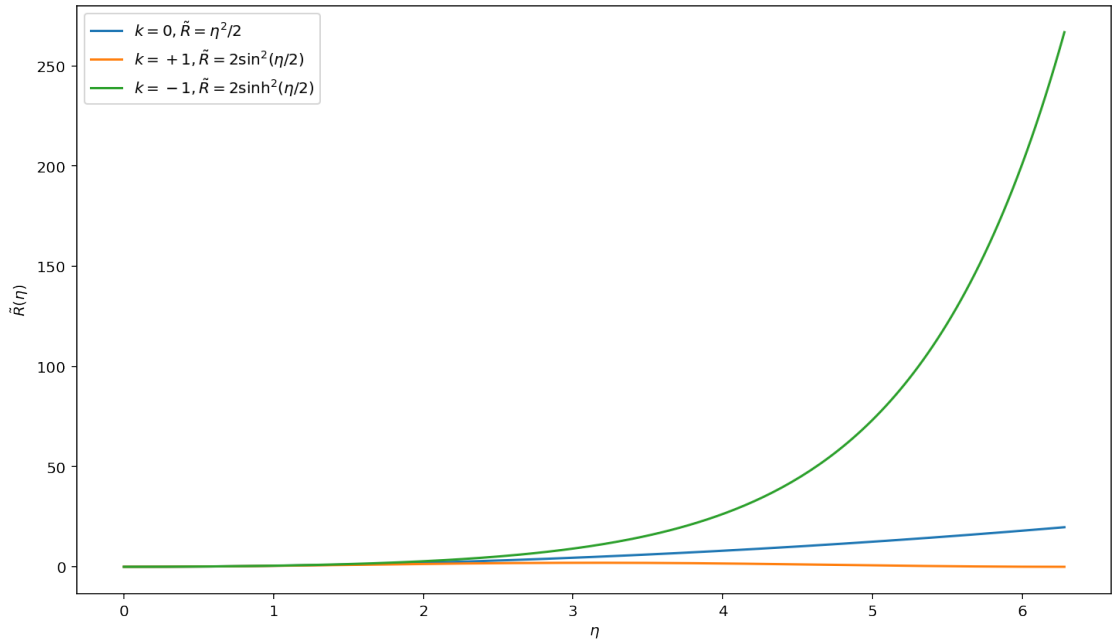
# \tilde{R} exact solutions
Rfuncz = lambda e: np.square(e) / 2
Rfuncp = lambda e: 2 * np.square(np.sin(e/2))
Rfuncn = lambda e: 2 * np.square(np.sinh(e/2))

fig = plt.figure()
plt.xlabel(r'$\eta$')
plt.ylabel(r'$\tilde{R}(\eta)$')

plt.plot(eta, Rfuncz(eta), label=r'$k=0, \tilde{R}=\eta^2/2$')
plt.plot(eta, Rfuncp(eta), label=r'$k=+1, \tilde{R}=2\sin^2(\eta/2)$')
plt.plot(eta, Rfuncn(eta), label=r'$k=-1, \tilde{R}=2\sinh^2(\eta/2)$')
plt.legend(loc="upper left")
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<matplotlib.legend.Legend at 0x7fac786d8e80>
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# exact R from \tilde{R}
Rz = lambda e: np.sqrt((3*c**2) / (4*G*pi*p*Rfuncz(e)))
Rp = lambda e: np.sqrt((3*c**2) / (4*G*pi*p*Rfuncp(e)))
Rn = lambda e: np.sqrt((3*c**2) / (4*G*pi*p*Rfuncn(e)))

# exact R^* from R
Rstarz = lambda e: 4*G*pi*p*Rz(e)**3/(3*c**2)
Rstarp = lambda e: 4*G*pi*p*Rp(e)**3/(3*c**2)
Rstarn = lambda e: 4*G*pi*p*Rn(e)**3/(3*c**2)

timez = lambda t,e: Rfuncz(e) # (c/R^*) dt / d\eta = \tilde{R}
timep = lambda t,e: Rfuncp(e)
timen = lambda t,e: Rfuncn(e)

ntz = odeint(timez, 0, eta) / (2*pi) # approximate t(eta) / 2\pi
ntp = odeint(timep, 0, eta) / (2*pi)
ntn = odeint(timen, 0, eta) / (2*pi)

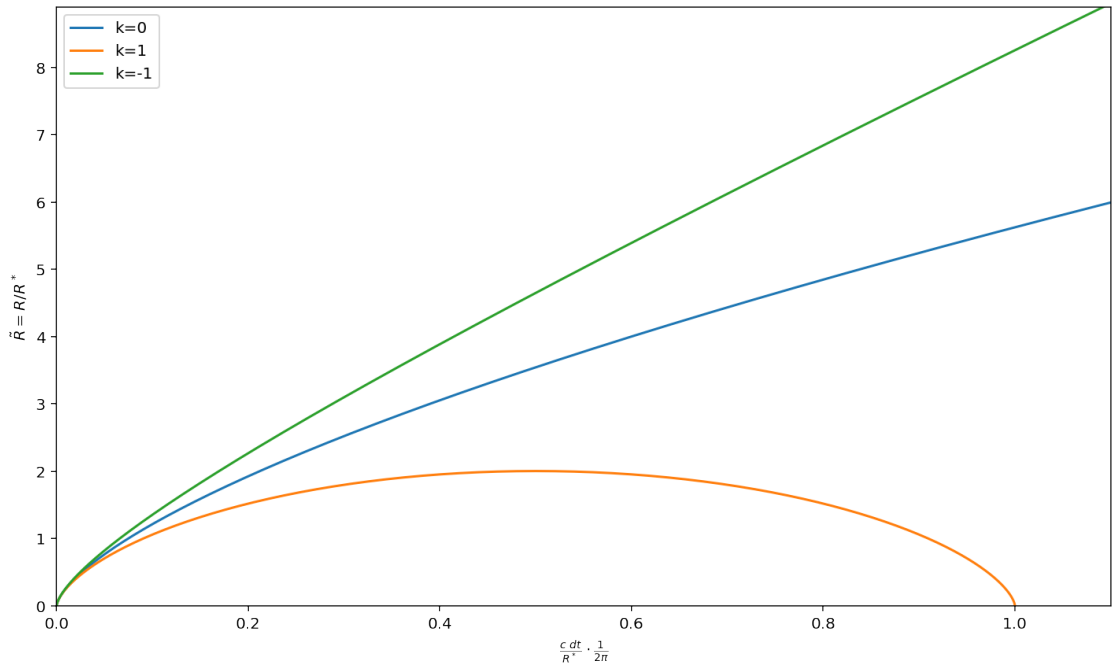
fig = plt.figure()
plt.plot(ntz, Rfuncz(eta), label="k=0")
plt.plot(ntp, Rfuncp(eta), label="k=1")
plt.plot(ntn, Rfuncn(eta), label="k=-1")

plt.xlim(0,1.1)
plt.ylim(0,8.9)
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plt.xlabel(r'$\frac{c~dt}{R^*}\cdot \frac{1}{2\pi}$')
plt.ylabel(r'$\tilde{R}=R/R^*$')
plt.legend(loc="upper left")
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[4]: <matplotlib.legend.Legend at 0x7fac7037d880>

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## 3--4)

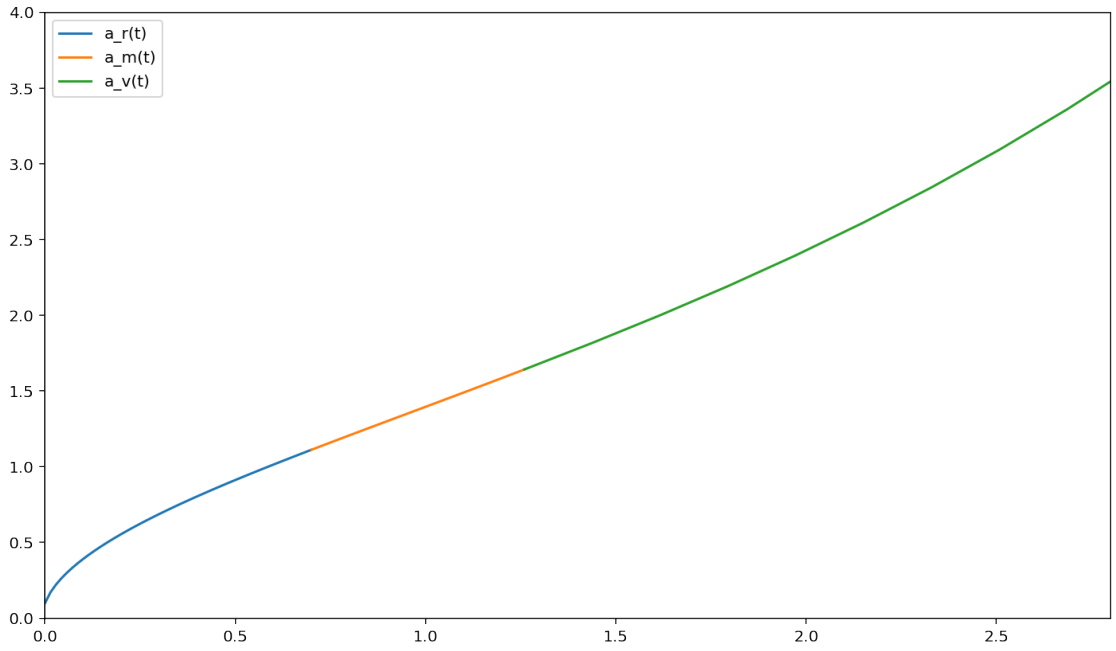
rmequil = 0.7
rvequil = 2**(1/3)

tt = np.linspace(0,rmequil)
tm = np.linspace(rmequil, rvequil)
tv = np.linspace(rvequil, 10)
dadt = lambda a,t: sqrt(0.4/a + 0.4/(a**2) + 0.2*a**2)
asol = odeint(dadt, 0.1, tt)
am = odeint(dadt, asol[-1], tm)
av = odeint(dadt, am[-1], tv)
plt.plot(tt, asol, label="a_r(t)")
plt.plot(tm, am, label="a_m(t)")
plt.plot(tv, av, label="a_v(t)")
# plt.plot(tt, a*tt**(1/2), label="t^{1/2}")
# plt.plot(tt, b*tt**(2/3), label="t^{2/3}")
# plt.plot(tt, c*np.exp(d*tt), label="exp(t)")
plt.xlim(0,2.8)
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plt.ylim(0,4)
plt.legend(loc="upper left")
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[5]: <matplotlib.legend.Legend at 0x7fac6f76d8e0>

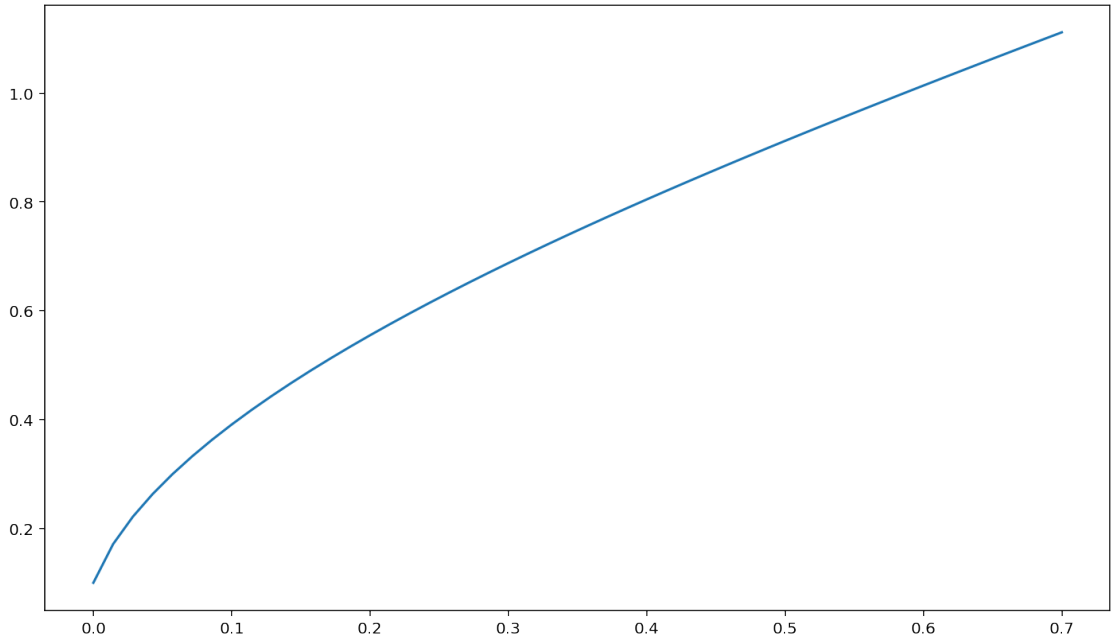
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plt.plot(tt, asol, label="a_r(t)")
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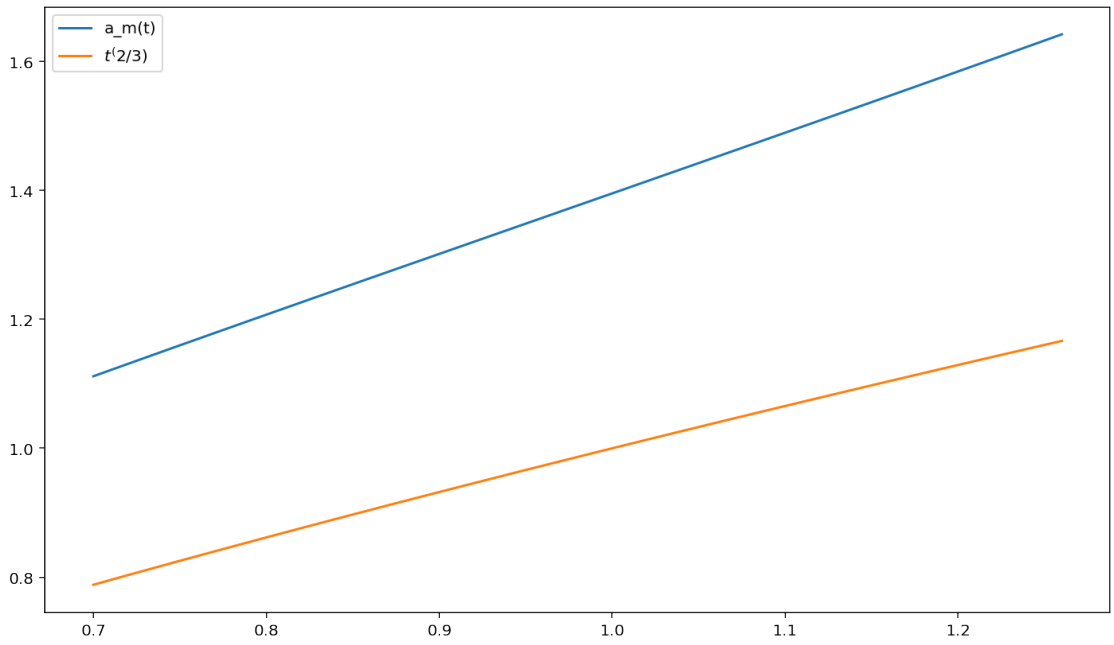
[6]: [<matplotlib.lines.Line2D at 0x7fac6f6d9a30>]

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plt.plot(tm, am, label="a_m(t)")
plt.plot(tm, tm**(2/3), label=r'$t^(2/3)$')
plt.legend(loc="upper left")
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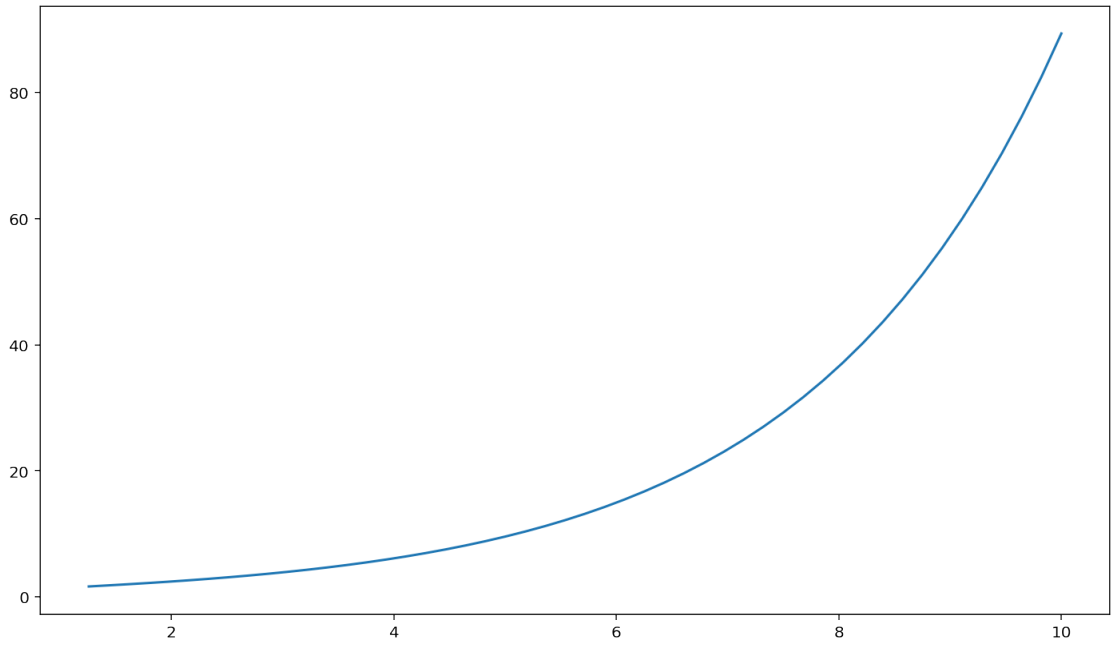
[7]: <matplotlib.legend.Legend at 0x7fac7035a100>

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plt.plot(tv, av, label="a_v(t)")
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[<matplotlib.lines.Line2D at 0x7fac7033ea90>]
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